

Lesson 1 Ros System Introduction

1. Ros System Introduction

Before the introduction to Ros system, we need to understand the definition of operating system. An operating system provides some software and hardware interfaces that we can use directly. Hence, different platform and different function require different operating system to complete the underlying operation.

Ros system is applicable to open-source robot operating system. It offers services similar to operating system, including Hardware abstract description, bottom driver management, shared functions execution, message transfer between programs, program distribution package management. It also provides some tools and libraries for you to obtain, build, write and execute program of multi-machine integration program.

In addition, the biggest difference between Ros and other operating system is interprocess communication. Interprocess communication of traditional operating system is called by system, which consumes lots of system resources and is not suitable for robot.

A robot consists of multiple sensors and perform the specific task through the communication and collaboration of each module. Hence, to be a robot, effective and quick interprocess communication is essential and Ros system offer such function.

2. Features of Ros

- ① Node to node. Separate function.

- ② Simple structure and high integration.
- ③ Abundant toolkits and support multi-language.